

UST'-KULOM, Russian Federation

WMO: 238030

Lat: 61.6994N

Lon: 53.6950E

Elev: 470

StdP: 14.45

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.1	-24.2	-36.3	0.7	-31.2	-29.2	1.1	-23.9	18.4	19.6	16.0	19.3	2.3	90	0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	82.6	67.9	78.9	65.7	75.3	63.7	70.0	79.4	67.6	76.2	65.4	73.1	5.2	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.5	99.3	74.7	64.3	91.9	72.8	62.1	84.8	70.1	34.2	79.6	32.3	76.3	30.5	73.0	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	14.6	12.3	-33.0	87.4	8.1	3.2	-38.9	89.7	-43.6	91.6	-48.2	93.4	-54.1	95.8
			-34.4	72.7	6.7	3.3	-39.2	75.1	-43.1	77.0	-46.8	78.8	-51.6	81.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.4	4.3	6.8	20.4	34.1	46.6	56.4	62.6	55.9	46.7	34.3	20.7	10.6
	DBStd	22.47	15.24	14.17	9.93	9.04	9.70	8.37	7.91	7.46	6.88	7.51	11.98	16.68
	HDD50	7008	1418	1211	917	482	186	30	7	25	143	488	878	1223
	HDD65	11632	1883	1631	1382	928	573	282	140	295	550	952	1328	1688
	CDD50	957	0	0	0	4	82	221	397	209	43	1	0	0
	CDD65	105	0	0	0	0	3	22	65	15	0	0	0	0
	CDH74	1012	0	0	0	1	58	201	622	129	1	0	0	0
	CDH80	214	0	0	0	0	8	37	142	27	0	0	0	0
Wind	WSAvg	5.6	5.6	5.6	6.2	6.2	6.1	5.4	4.4	4.7	5.0	6.0	5.9	5.9
Precipitation	PrecAvg	24.4	1.7	1.3	1.3	1.3	2.0	2.8	2.4	2.7	2.3	2.4	2.2	1.7
	PrecMax	30.7	3.0	3.0	2.2	2.9	4.6	6.1	4.6	5.7	4.5	4.0	4.4	3.1
	PrecMin	19.5	0.6	0.2	0.3	0.0	0.9	0.6	1.0	0.9	0.6	0.9	0.5	0.7
	PrecStd	2.8	0.6	0.7	0.5	0.7	1.0	1.4	0.9	1.3	1.0	0.9	0.9	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.8	34.3	44.2	66.4	80.0	84.6	87.1	83.6	69.2	58.6	45.0	36.1
		MCWB	31.9	32.4	36.8	51.4	60.4	68.8	70.4	69.9	60.4	52.2	43.7	35.0
	2%	DB	30.7	32.0	39.0	56.4	74.1	79.9	84.0	77.3	64.0	51.6	38.9	33.8
		MCWB	29.9	30.9	33.4	45.4	57.8	66.5	69.4	65.2	56.7	48.4	37.7	32.8
	5%	DB	28.1	29.1	36.0	51.1	68.8	75.6	80.8	72.8	60.6	47.7	35.5	32.0
		MCWB	27.5	28.0	32.4	42.5	55.5	63.8	67.1	63.0	54.8	45.5	34.7	31.3
	10%	DB	24.1	25.6	33.4	47.0	63.8	71.7	77.3	68.6	57.2	45.0	33.4	29.7
		MCWB	23.5	24.4	30.8	39.6	52.3	61.0	65.3	60.9	53.1	43.1	32.6	28.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.2	32.9	37.9	53.0	63.3	71.4	74.0	71.8	61.4	53.4	43.6	35.3
		MCDB	32.5	33.7	42.6	65.3	76.6	80.9	83.6	81.5	66.8	56.1	45.2	35.5
	2%	WB	29.9	30.6	34.8	47.1	59.4	68.1	70.8	67.1	58.0	48.5	37.8	33.0
		MCDB	30.6	31.7	37.5	54.2	70.5	77.6	80.6	75.1	62.3	50.9	38.7	33.6
	5%	WB	27.4	27.9	32.9	43.3	56.7	65.5	68.9	64.2	55.7	45.9	34.8	31.3
		MCDB	27.9	29.1	35.4	49.7	67.2	74.0	78.0	70.5	59.2	47.6	35.4	32.0
	10%	WB	23.5	24.6	31.0	40.5	53.7	62.7	66.8	61.8	53.5	43.1	32.8	28.9
		MCDB	24.1	25.5	33.5	45.8	62.3	69.6	74.6	67.5	57.0	44.7	33.3	29.7
Mean Daily Temperature Range	5% DB	MDBR	11.4	13.4	15.2	15.6	18.1	17.9	18.8	15.9	13.1	8.0	8.0	10.3
		MCDBR	11.2	10.2	14.6	23.3	26.3	22.9	24.5	22.2	19.6	12.4	7.3	6.9
		MCWBR	11.2	9.3	11.0	14.9	14.5	12.4	11.7	11.8	12.7	9.8	7.2	6.9
	5% WB	MCDBR	11.2	9.5	12.6	20.5	23.9	20.8	22.2	20.0	17.0	11.9	7.3	6.9
		MCWBR	11.2	9.2	10.1	14.3	14.9	12.7	12.1	11.9	12.0	9.8	7.3	6.9
Clear-Sky Solar Irradiance	taub	0.258	0.300	0.339	0.367	0.364	0.366	0.404	0.385	0.345	0.319	0.267	0.236	
	taud	1.962	2.032	2.018	2.120	2.396	2.440	2.332	2.368	2.448	2.335	2.059	1.921	
	Ebn at Noon	126	200	236	257	272	273	257	252	243	201	124	77	
	Edh at Noon	13	25	36	40	33	32	35	31	24	18	12	8	
All-Sky Solar Radiation	RadAvg	88	316	710	1156	1532	1614	1662	1140	651	254	84	41	
	RadStd	12	66	104	156	152	123	141	113	85	28	20	6	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page